

Enhancing Travel Planning with LLMs and Machine Learning Systems

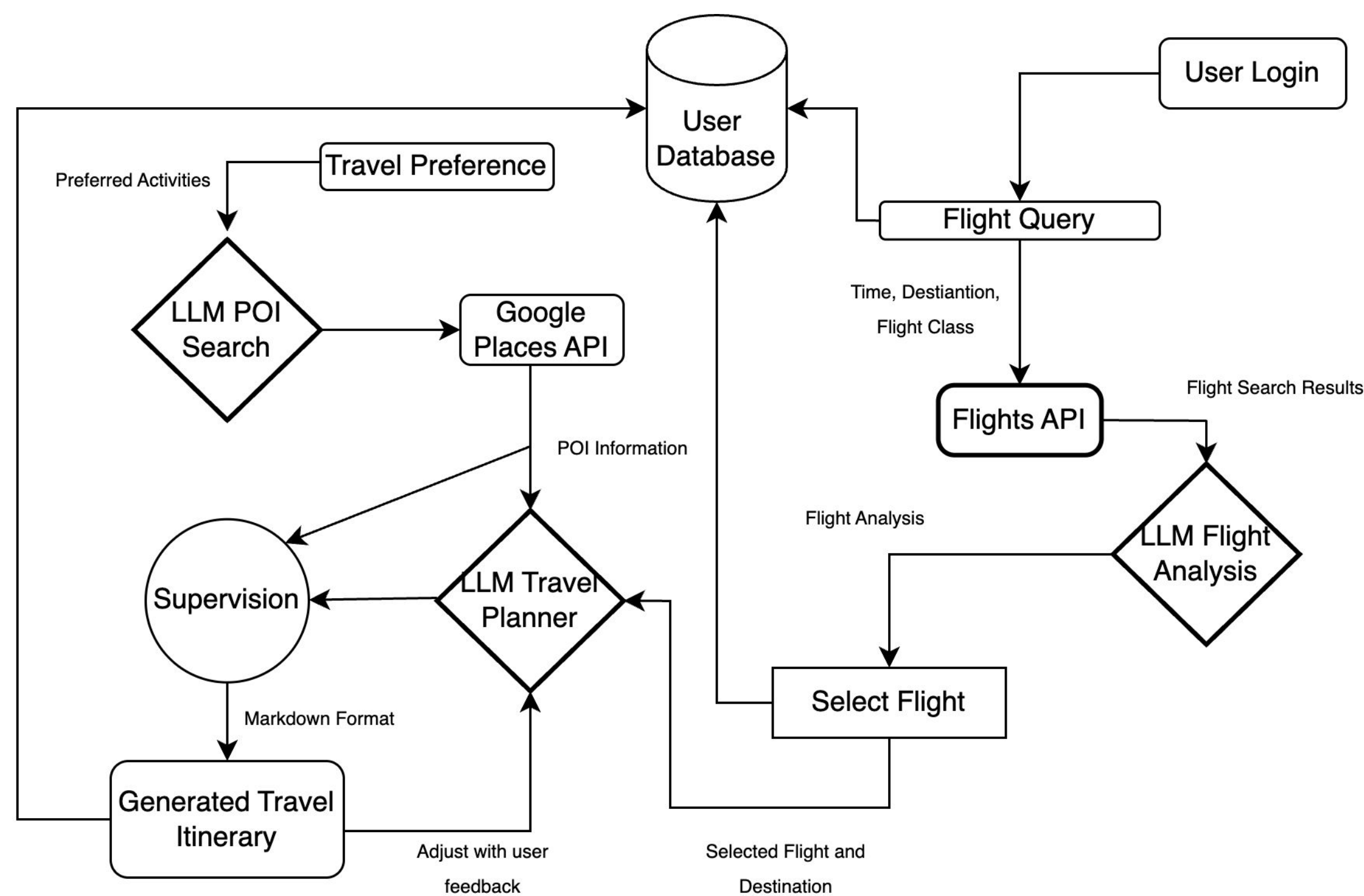
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Introduction and Objective

As making travel plans can often feel like a toil—with people spending hours watching countless travel videos and reading numerous blogs, only to find themselves surrounded by mostly irrelevant information that doesn't align with their preferences—our travel planning system offers a smarter solution. It integrates **LLM-powered itinerary creation** with **dual-approach recommendation algorithms**, combining **flight data** and **points of interest (POI)** information to craft personalized travel experiences. By leveraging **user-based** and **content-based filtering**, the system effectively addresses information overload and simplifies trip organization.

We also implemented **clustering analysis** to group users based on their travel behaviors and preferences, allowing for even more targeted recommendations. This approach enhances personalization without requiring sensitive personal data, further protecting user privacy.

Program Structure and User Journey



The flowchart above outlines our system, where each user is assigned a **unique ID** stored in the **User Database** to personalize their experience. After **login**, users submit a **flight query**, which is processed via the **Flights API** and analyzed by the **LLM Flight Analysis** to help select the best option.

Once a destination is chosen, **travel preferences** are used by the **LLM POI Search** and **Google Places API** to gather **points of interest**. The **LLM Travel Planner** then generates a **personalized itinerary**, which can be refined with **user feedback** for improved accuracy.

Recommendation System

Our recommendation system offers personalized destination suggestions through two approaches. **Content-based filtering** clusters users by preferences (e.g., activities, food, cabin type) and uses XGBoost to predict suitable destinations—ideal for new users. **User-based filtering** matches users with similar historical profiles and recommends destinations enjoyed by their closest behavioral neighbors, making it effective for returning users. Together, they balance cold-start adaptability with long-term personalization.

Additionally, **the system predicts user interests based on selected destinations**. By estimating destination-cluster probabilities and training cluster-specific Random Forest models, we generate weighted predictions of activity preferences.

Part 1: Destination Recommendation

Method 1: Content-Based Filtering (Cold Start)

- **Key Idea:** Match user profiles via interests and cluster-based modeling
- **Pipeline:**
 - K-Means clusters user profiles
 - Train XGBoost per cluster
 - For new users: assign cluster → predict destination (result: *Cluster 1 Accuracy: 93%*)

Method 2: User-Based Filtering (Warm Start)

- **Key Idea:** Recommend destinations liked by similar users
- **Application:** Exploratory and history-based personalization
- **Pipeline:**
 - Aggregate average user profile
 - Find nearest neighbor (NN)
 - Recommend NN's favorite destination closest to user profile

Part 2: Predict User Preferences Based on Destination

- **Key Idea:** infers a user's activity preferences based solely on their selected destination by combining destination-cluster probabilities with cluster-specific Random Forest models
- **Pipeline:**
 - Compute the destination-cluster mapping by estimating the empirical probability

$$\mathbb{P}(C = c \mid d) = \frac{\mathbb{P}(d \mid C = c) \cdot \mathbb{P}(C = c)}{\mathbb{P}(d)}$$

- Train Random Forest and gives weighted prediction

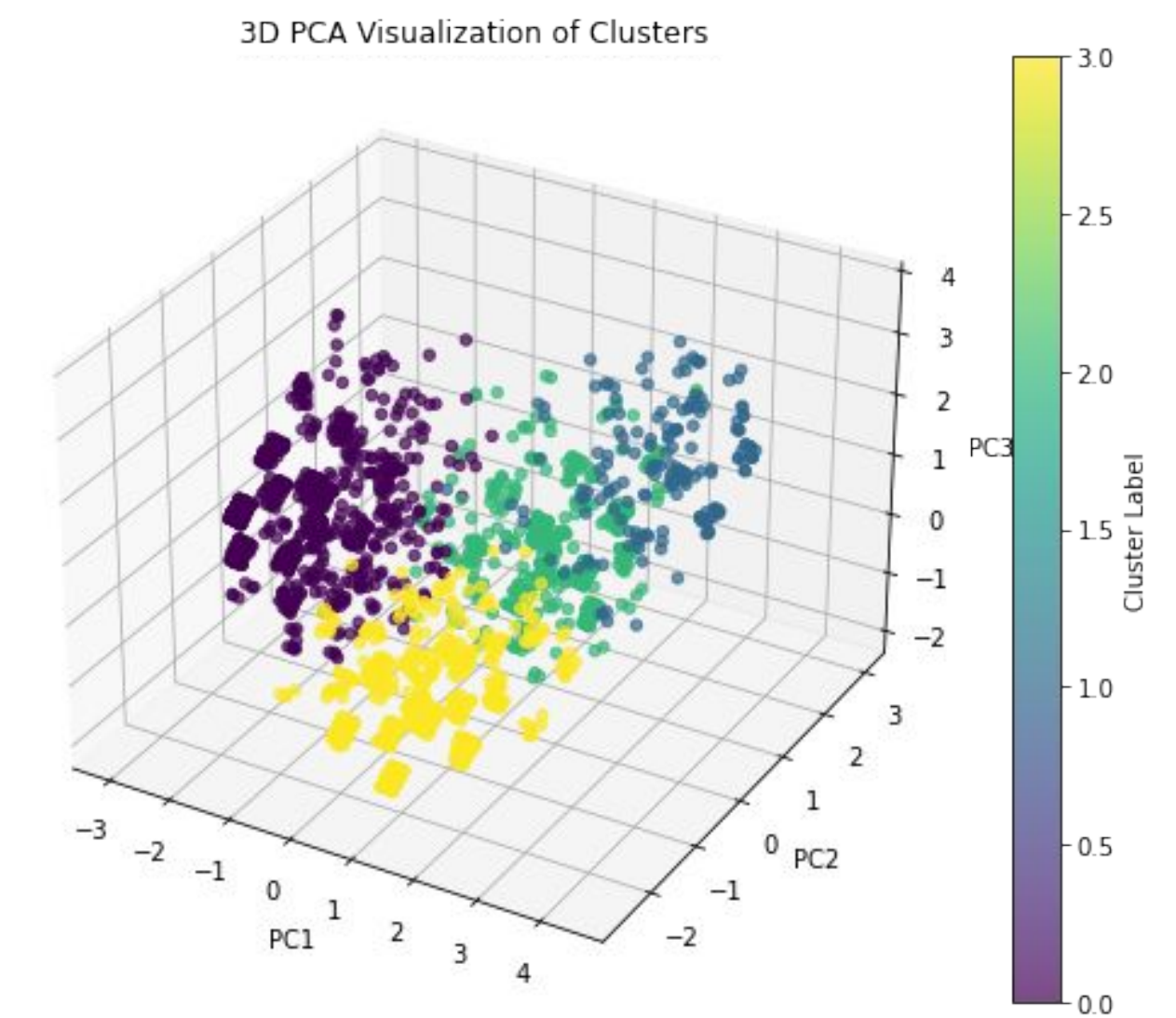
$$\hat{y}_i = \sum_{c=1} \mathbb{P}(C = c \mid d) \cdot f_{c,i}(d, c)$$

User preferences → XGBoost / Nearest Neighbor → Destination recommendation

Destination only → Cluster-weighted Random Forest → Inferred preferences

Results

Cluster	Frequent Destinations	Preferences
0	San Francisco, CA Seattle, WA Chicago, IL	Divergent food preferences and activity interests
1	Los Angeles, CA	Highly engaged in nature/outdoors, entertainment, shopping, recreation, and nightlife
2	Miami, FL Orlando, FL	High engagement in nature/outdoors, entertainment, and recreation
3	New York City, NY Washington, D.C. Boston, MA Denver, CO	High interest in cultural activities, nature/outdoors, shopping, and nightlife. Strong food preferences for European, Middle Eastern, and Caribbean cuisines.



To visualize our clustering results, we applied **PCA** to our dataset, reduced it to the top three principal components, and plotted them in 3D. The data points were grouped into four clusters for better interpretability. For example, Cluster 0 contains travelers who frequently visit destinations such as San Francisco, CA, and Seattle, WA, and exhibit divergent food preferences and activity interests. Clustering user profiles and providing tailored recommendations based on the common activities and preferences of their group is an effective method to enhance personalization and improve user satisfaction in travel planning.

Summary and Next Steps

To equip our travel planning system with more robustness, integrating the following APIs might further our program's accuracy and personalization:

- Incorporate **weather APIs** to tailor activity based on real-time weather conditions
- Add **local event APIs** to recommend festivals or seasonal activities.

Regarding our user profiling and clustering recommendation analysis, although we incorporated a generated test dataset for sheer analysis purposes without access to corporate user information backends, the clustering approach demonstrates the potential to segment real-world users based on their idiosyncratic travel behaviors and preferences. This method can be scaled and applied to actual user data to provide meaningful insights and deliver personalized recommendations or potential promotions.

Our proposed approach also enhances user privacy, as it does not collect nor rely on sensitive personal information such as income, gender, or age group. Instead, our study's main focus is solely on travel behaviors and preferences.

References

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- [3] Permana, Kurniawan Eka, et al. Tourism Recommendation System Using User-Based Collaborative Filtering. 2025. AIP Conference Proceedings, <https://doi.org/10.1063/5.0241251>